

Learning Stochastic Image Transformations: Synthetic Static, Generalization, and Reversible Transport

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Abstract

We compare independently learned image transformations with networks possessing an explicit inverse, using 809 Pokémon sprites and synthetic uniform static. The experiments distinguish recovery guaranteed by architecture from learning a prescribed stochastic transformation. We study discrete modular noise and a Gaussian diffusion endpoint, explicitly accounting for conditioning information and serialized payloads. Results are labeled *Frozen test evaluation*. This study concerns generalization and numerical transport, not cryptographic security. Uniform static is evaluated as a training distribution, not assumed to cover the distribution of structured images. Across 35 short training configurations, validation selected a 44,296-parameter Gaussian coupling model trained on sprites alone. Its held-out forward-plus-reverse normalized mean absolute error was 0.04924, compared with 0.05155 for equal mixed training and 0.12187 for static-only training. Thus uniform static did not improve target learning under the tested fixed update budget. Retained byte coupling models achieved exact serialized self-round trips across all evaluated test domains, although agreement with the prescribed byte transformation remained imperfect.

1 Motivation and prior experiment

The earlier implementation trained and evaluated on the same 809 images, preserved alpha masks, and used a 1,000-step Gaussian schedule with signal coefficient approximately 5.07×10^{-12} before clipping and 8-bit quantization. These conditions cannot establish general recovery of unseen RGB content. We retain that process as a diagnostic and evaluate a silhouette-only nearest-training-mask retrieval baseline. This baseline quantifies a shortcut; it is not a reconstruction model trained on the test images.

2 Related approaches

Real NVP introduced tractable invertible continuous coupling transformations [1]. Discrete flows and integer discrete flows provide corresponding constructions for discrete domains [2, 3]; continuous invertibility alone does not guarantee recovery following quantization. Efficient restoration architectures such as NAFNet [4] motivate compact convolutional comparisons, although our baseline is a plain residual CNN, not a reproduction of NAFNet. Adversarial neural cryptography [5] addresses a different objective, with keys and an attacker. We do not infer confidentiality from noise appearance, reconstruction accuracy, or invertibility.

3 Methods

For byte tensors, the teacher is $y = (x + n) \bmod 256$, with independent uniform integer noise and inverse $x = (y - n) \bmod 256$. Repeated independent modular additions collapse into one cumulative field, so we do not pretend that increasing the number of such steps increases expressive power. For normalized floating-point tensors the teacher is $y = ax + n$, $a = 0.5$, with $n \sim \mathcal{N}(0, 1 - a^2)$. This is exactly the endpoint distribution of 16 Gaussian updates with $\beta = 1 - a^{2/16}$. The inverse is $(y - n)/a$. Both learner directions receive the realized noise field. The analytic inverse requires no training; learned models are studied as controlled approximation experiments, not as more efficient substitutes for this elementary arithmetic. Reported conditioning sizes count the full field. A compact seeded alternative would additionally require the generator version and stream position; the eight-byte seed field alone is not a complete transport specification.

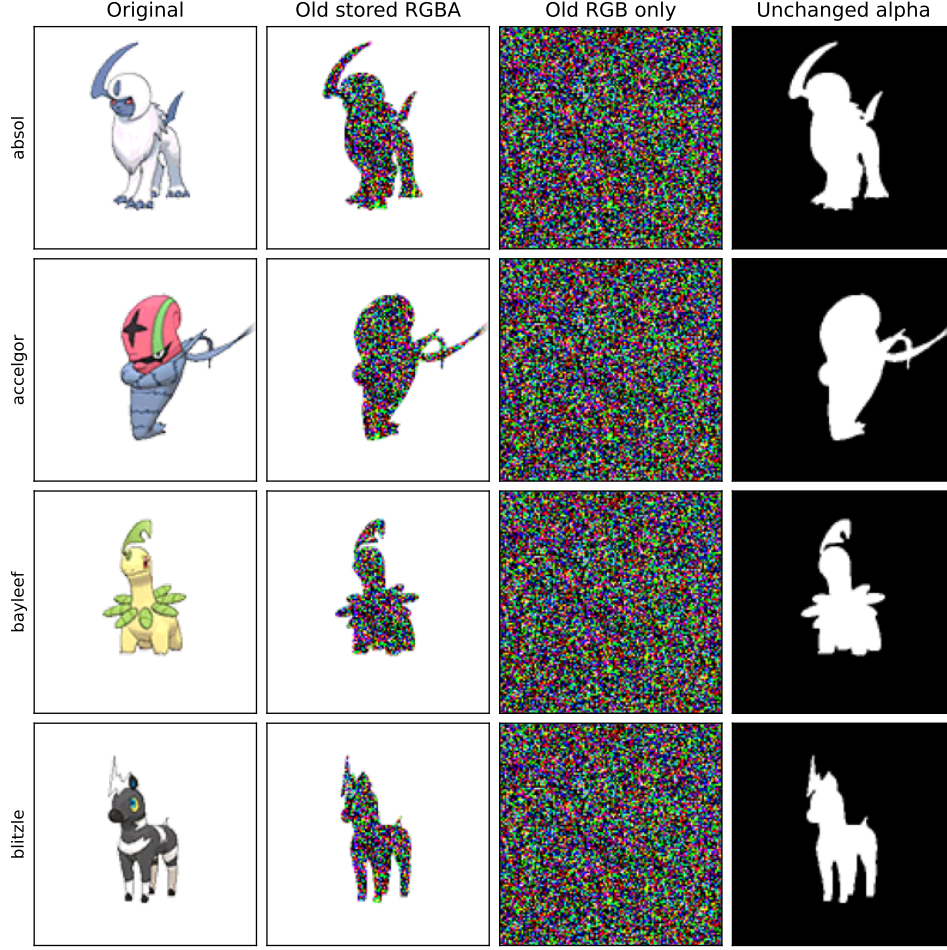


Figure 1: The first four validation images in manifest order. Under the old process, the RGB fields are identical; unchanged alpha carries the distinct silhouettes. Showing only the stored RGBA render can conceal this failure.

The independent model pair consists of six residual blocks per direction with 32 channels and no downsampling or batch normalization. Byte outputs use four 256-category heads; float outputs use four regression channels. Coupling models use four alternating half-channel updates, each conditioned on the unchanged half and the full noise field through a small convolutional network. Integer shifts use straight-through rounding for training and hard rounded values for inference. The Gaussian model uses fixed coupling scales whose product matches the teacher’s attenuation. This is an explicit inductive bias favoring that architecture. No spatial mixing teacher is included in this first campaign. For a byte coupling block, writing the channel halves as (x_A, x_B) , we use

$$y_A = x_A, \quad y_B = (x_B + Q(t_\theta(x_A, n))) \bmod 256. \quad (1)$$

Its inverse subtracts the identical rounded shift, recomputed from y_A and n . Here Q rounds to the nearest integer with ties to even. The continuous block instead uses $y_B = sx_B + t_\theta(x_A, n)$ with nonzero fixed s , inverted by subtraction and division. Blocks are inverted in reverse order. Thus invertibility holds independently of whether the parameters match the prescribed teacher, subject to reproducing the numerical conditioner outputs.

Byte training uses cross entropy for independent maps and circular byte distance for coupling maps; Gaussian training uses mean squared error. An exploratory byte follow-up, motivated by poor endpoint learning in the initial trials, keeps the coupling architecture unchanged and adds intermediate teacher supervision. For each channel half, integer shifts $\lfloor n/2 \rfloor$ and $n - \lfloor n/2 \rfloor$ sum to the teacher field. An auxiliary mean squared error on shifts divided by 128 is averaged across layers and added to each direction’s loss. Intermediate inputs are teacher-forced. This intervention exposes additional teacher information and is not an architecture-only comparison. AdamW uses learning rate 3×10^{-4} , weight decay 10^{-5} and gradient clipping at 1. Effective batch size is 16. Discrete coupling evaluation uses FP32 without TF32; integer recovery depends on reproducing the same rounded conditioner outputs. Cross-device or cross-library bitwise compatibility is not established here.

4 Experimental protocol

All images retain native 120×120 resolution. Alpha is transformed rather than exposed as a silhouette. Approximately 80/10/10 percent of images are split before transformation, grouping identical pixels, named variants, and available CSV evolution links. The supplied links are incomplete, so this is not a complete evolutionary-family holdout. Synthetic samples use independent uniform bytes in all four channels. Training regimes are sprite only, static only, and equal mixed batches. Evaluation also includes separate static samples and structured probes. The frozen manifest contains 647 training, 81 validation, and 81 test images.

One epoch is defined as 100 optimizer updates. Initial screening compares 12 configurations, followed by three guided-training byte configurations at the same update budget, for at most 10 epochs; promoted architectures receive at most 100 epochs for each regime and seeds 17, 29, and 43. Hardware profiling determines a shared update cap within each phase. Fixed-noise diagnostics use the same budget as their fresh-noise counterparts. Validation alone determines checkpoints and promotion, averaging over data regimes. Float round-trip errors below 10^{-6} are treated as tied. The executed campaign comprises 35 trained configurations: 15 screening runs at 1,000 updates, 18 repeated runs at 2,000 updates, and two fixed-noise diagnostics at 2,000 updates. These are 10 and 20 defined epochs, respectively. Byte ranking first prioritizes exact full-image round trips, then forward and reverse target-byte accuracy, and finally round-trip MAE. Gaussian ranking first uses round-trip MAE above the tolerance, then summed forward and reverse target MAE. The main paper track is chosen by relative validation target-error improvement over its untrained control, with absolute errors and transport formats reported separately; this is not a security ranking. All retained checkpoints are hashed and frozen before test access. The same test images are paired across conditions; seed standard deviations describe training variation and are not population confidence intervals. After selection, fixed- and fresh-noise-trained mixed models are cross-evaluated on both fixed and fresh validation fields. This separates performance on a known realization from transfer to new fields without using the test set for tuning.

5 Results

5.1 Architecture and training screening

The screening table contains single-seed validation results at 1,000 updates per configuration. Target MAE sum combines forward and reverse teacher-endpoint errors. The guided variant uses additional intermediate supervision; comparisons with the original coupling loss isolate that intervention without increasing model size.

Track	Model	Data	Target MAE sum	Round-trip MAE	Target byte %
byte	coupling	mixed	0.69	0	0.6
byte	coupling	pokemon	0.6912	0	0.6
byte	coupling	static	0.6893	0	0.6
byte	guided	mixed	0.4093	0	16.4
byte	guided	pokemon	0.3752	0	15.2
byte	guided	static	0.4402	0	9.3
byte	pair	mixed	0.1987	0.1074	5.0
byte	pair	pokemon	0.2021	0.1074	4.7
byte	pair	static	0.8194	0.4917	0.4
gaussian	coupling	mixed	0.08548	7.651e-08	—
gaussian	coupling	pokemon	0.08173	7.583e-08	—
gaussian	coupling	static	0.1428	7.606e-08	—
gaussian	pair	mixed	0.123	0.0643	—
gaussian	pair	pokemon	0.08151	0.04099	—
gaussian	pair	static	0.6099	0.2545	—

5.2 Retained approaches

The table reports mean normalized errors, with byte errors divided by 255. Forward error measures agreement with the teacher; reverse error uses teacher outputs; round-trip error uses the learner’s own serialized output. These are different questions. Exact image recovery and byte accuracy, per-domain results, untrained controls, hardware costs, and per-seed values accompany the paper as machine-readable artifacts. Full-image errors include alpha and background RGB values. Foreground-only RGB errors, using the original nonzero alpha mask, are recorded separately so that background-heavy sprites do not conceal errors in visible content.

Track	Model	Data	Forward MAE	Reverse MAE	Round-trip MAE
byte	guided	mixed	0.01822	0.3421	0
byte	guided	pokemon	0.01661	0.3281	0
byte	guided	static	0.01915	0.4257	0
gaussian	coupling	mixed	0.01718	0.03437	7.685e-08
gaussian	coupling	pokemon	0.01641	0.03283	7.583e-08
gaussian	coupling	static	0.04153	0.08034	8.194e-08

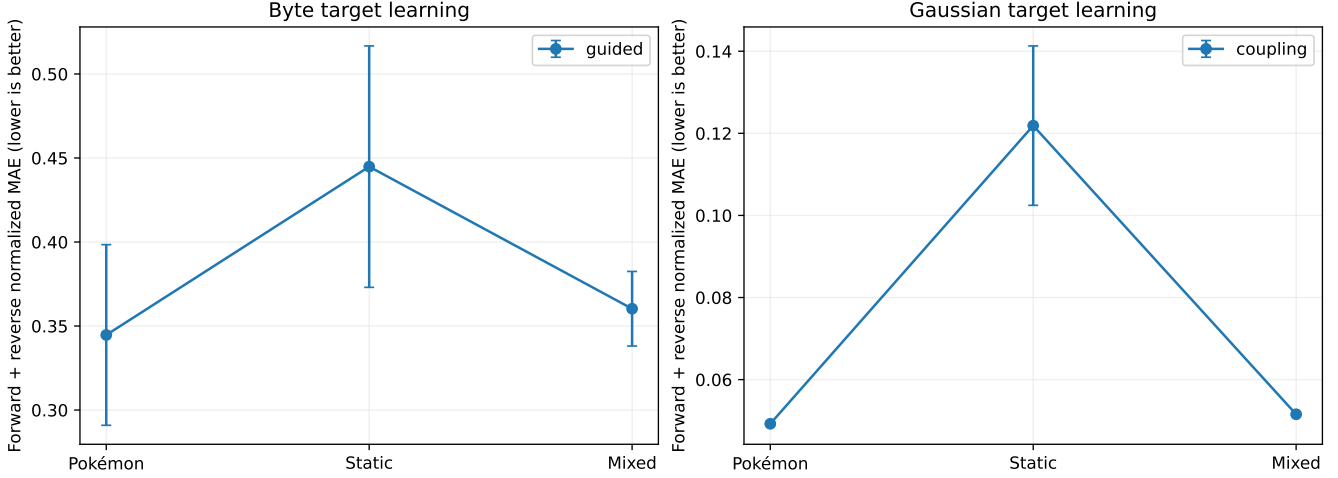


Figure 2: Target transformation learning across data regimes. Error bars, when available, show sample standard deviation across three training seeds.

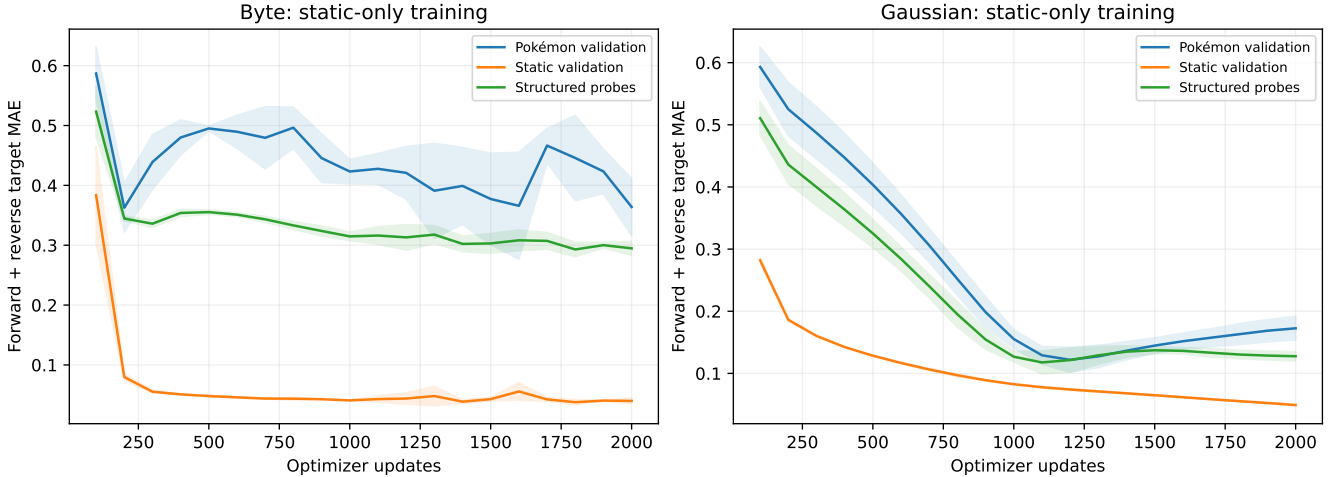


Figure 3: Validation trajectories during static-only training of retained models. Lines average the training seeds; bands show sample standard deviation. Static, sprite, and structured-probe errors are reported separately to expose distribution transfer failures. These are validation trajectories, not repeated test queries.

5.3 Selection and transfer

Validation selected the coupling architecture with pokemon training for the gaussian track as the main approach. This selection was frozen before test evaluation. Both transformation tracks and all training regimes remain reported.

For byte, the mean held-out forward-plus-reverse MAE was 0.34467 with sprite training, 0.44486 with static training, and 0.36029 with mixed training. These descriptive differences concern target learning; exact self-inversion

alone is not evidence of transfer.

Byte forward-target accuracy (mean and sample standard deviation across training seeds) was pokémon: $20.88 \pm 0.70\%$; static: $10.59 \pm 0.96\%$; mixed: $19.23 \pm 1.01\%$. These percentages measure agreement with the prescribed teacher, not self-round-trip recovery.

For gaussian, the mean held-out forward-plus-reverse MAE was 0.049237 with sprite training, 0.12187 with static training, and 0.051555 with mixed training. These descriptive differences concern target learning; exact self-inversion alone is not evidence of transfer.

The largest Gaussian learned-round-trip error across retained test runs and domains was $1.8477\text{e-}06$. This is comfortably below half an 8-bit step ($1/510$), including input float-rounding error, so rounding reconstructed values to the nearest original byte recovers the input byte grid. This is a numerical recovery property, not evidence of target learning.

5.4 Costs and diagnostics

Extended runs used 2000–2000 updates (20–20 defined epochs), with an effective batch size of 16. The retained coupling networks have 44,296 parameters; the independent byte and Gaussian pairs have 294,208 and 226,888 parameters respectively.

The original process yielded 1 unique RGB arrays among 81 validation images, while preserving alpha exactly. Silhouette-only nearest-training-mask retrieval achieved normalized RGB MAE 0.074125. Clipping and quantizing the gentler Gaussian endpoint to PNG produced analytic recovery MAE 0.77363; this illustrates why Gaussian results require an unclipped float payload.

5.5 Noise realization transfer

For byte, the fresh-noise-trained mixed model had validation target MAE sum 0.37334 on fresh fields and 0.3723 on the fixed field. The fixed-noise-trained counterpart had errors 0.36328 on fresh fields and 0.35988 on the fixed field. This single-seed diagnostic crosses training and evaluation noise conditions; it is excluded from architecture selection.

For gaussian, the fresh-noise-trained mixed model had validation target MAE sum 0.051844 on fresh fields and 0.051489 on the fixed field. The fixed-noise-trained counterpart had errors 0.059442 on fresh fields and 0.053802 on the fixed field. This single-seed diagnostic crosses training and evaluation noise conditions; it is excluded from architecture selection.

6 Limitations and interpretation

Exact self-inversion is a property of the coupling architecture, including before training. It does not prove the target was learned. Both teachers are elementary conditional transformations; results do not establish spatial, multimodal, or unseen-transformation generalization. Static-only validation selection still uses held-out Pokémon images, so the training distribution is synthetic but model selection is not fully blind to the target domain. Small sprite data, incomplete family metadata, differing losses and parameter counts, a single GPU, and three training seeds constrain conclusions. Independent pairs use teacher-endpoint supervision rather than joint cycle-consistency training; their results do not rule out improvements from that alternative. Architecture selection is restricted to the small candidates tested here, not a search for a globally optimal model. At equal update budgets, the mixed regime receives half as many Pokémon presentations as sprite-only training. This is a compute-budget comparison. An exposure-matched follow-up would retain the same number of real-image presentations and add synthetic updates, with the associated extra computation. An unrelated GPU workload was observed during the extended trials; later wall-clock training and inference times are descriptive shared-machine measurements, not isolated speed benchmarks. Architecture profiling preceded that workload. On the RTX 5070 (12 GB), profiling measured approximately 0.3–1.1 GB of peak PyTorch tensor allocation across candidates; this excludes driver and other process memory and is not total board usage. Gaussian float payloads are larger than byte tensors. Noise fields or reproducible seeds are explicit side information, and no secrecy, authentication, resistance to chosen-plaintext attacks, or deployment suitability is claimed. Further experiments should add spatial coupling teachers and independently test security objectives if pursued. To investigate the remaining distribution-transfer gap, a follow-up can compare white static with constant-color patches and spatially correlated synthetic textures, while retaining the same held-out evaluation discipline.

References

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